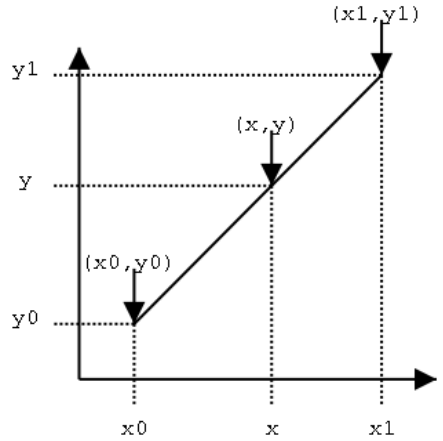
Interpolation and data fitting

Commonly, in typical experiments, we know $f(x_i)$ at various x_i, not necessarily equally spaced. Additionally, we may not quite know the analytical expression of $f(x)$ that describes the data and allow to calculate its value at arbitrary $x \neq x_i$.

If the desired x is/are between largest and smallest x_i's, the problem is of *interpolation*. If outside, then it is *extrapolation*.

Extrapolation is a terribly risky endeavor unless backed up by solid theoretical idea(s). Here we study only *interpolation*.

Interpolating (x, y) between (x_0, y_0) and (x_1, y_1) with a straight line,



$$\frac{y - y_0}{x - x_0} = \frac{y_1 - y_0}{x_1 - x_0} \Rightarrow y = y_0 + \frac{x - x_0}{x_1 - x_0}(y_1 - y_0)$$

Generalize this idea by considering a polynomial of degree N to approximate a function from a set of $N+1$ points

$$P_N(x_i) \equiv f(x_i) = y_i \quad \text{where } i = 0, 1, \ldots, N$$

The general form of $P_N(x)$ is

$$\begin{aligned}
P_N(x) &= a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1) + \cdots + \\
&\quad a_N(x - x_0)(x - x_1)(x - x_2)\ldots(x - x_{N-1}) \\
P_0(x_0) &= a_0 = y_0 = f(x_0) \\
P_1(x_1) &= a_0 + a_1(x_1 - x_0) = y_1 = f(x_1) \\
P_2(x_2) &= a_0 + a_1(x_2 - x_0) + a_2(x_2 - x_0)(x_2 - x_1) = y_2 = f(x_2) \quad \text{etc.}
\end{aligned}$$

For linear interpolation $N = 1$,

$$a_0 = y_0, \quad a_1 = \frac{y_1 - y_0}{x_1 - x_0} \Rightarrow P_1(x) = y(x) = y_0 + \frac{x - x_0}{x_1 - x_0}(y_1 - y_0)$$

A particularly useful form, although looks complicated, is

$$P_1(x) = y_0 - \frac{x - x_0}{x_1 - x_0} y_0 + \frac{x - x_0}{x_1 - x_0} y_1 = \frac{x - x_1}{x_0 - x_1} y_0 + \frac{x - x_0}{x_1 - x_0} y_1$$

$P_1(x)$ is the polynomial of degree 1 for $1 + 1$ data points.

Generalized interpolation formula by Lagrange

$$P_N(x) = \sum_{i=0}^{N} \prod_{k \neq i} \frac{x - x_k}{x_i - x_k} y_i$$

As an example, 2nd order polynomial for 3 data points x_0, x_1, x_2

$$P_2(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} y_0 + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} y_1 + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} y_2$$

Ex 1. Consider the following data table and show the $f(x = 4) = 20$

x	2	3	5	8	12
$f(x)$	10	15	25	40	60

Ex 2. To understand what is going on, take a smaller data set

x	0	10	20	30
$f(x)$	-250	0	50	-100

Each term in Lagrange formula *i.e.* coefficient of y_i

$$i=0 \qquad \frac{(x-10)(x-20)(x-30)}{(0-10)(0-20)(0-30)} = -\frac{x^3}{6000}+\frac{x^2}{100}-\frac{11x}{60}+1$$

$$i=1 \qquad \frac{(x-0)(x-20)(x-30)}{(10-0)(10-20)(10-30)} = \frac{x^3}{2000}-\frac{x^2}{40}+\frac{3x}{10}$$

$$i=2 \qquad \frac{(x-0)(x-10)(x-30)}{(20-0)(20-10)(20-30)} = -\frac{x^3}{2000}+\frac{x^2}{50}-\frac{3x}{20}$$

$$i=3 \qquad \frac{(x-0)(x-10)(x-20)}{(30-0)(30-10)(30-20)} = \frac{x^3}{6000}-\frac{x^2}{200}+\frac{x}{30}$$

Multiply each line with the corresponding $f(x_i)$ and add together the terms of like power to obtain

$$y(x) \equiv P_3(x) = -x^2 + 35x - 250$$

Lagrange's formula gives a simple quadratic description of the data.

If required, we can calculate, say, $f(x=15)=50$ either using the quadratic formula or directly from Lagrange's formula.

DIY : Find both way $f(x=1969)$ from the data

x	1951	1961	1971
$f(x)$	2.8	3.2	4.5

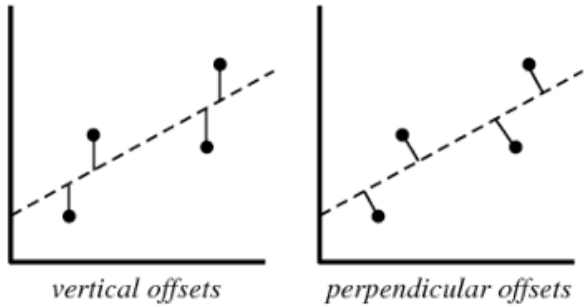
Least square fitting

Set of data points generated (x_i, y_i) are expected to be described by a **known** function $f(x)$ but with undetermined coefficients.

Examples – stress-strain data points within elastic limit, voltage-current data points. Task is to determine modulus of elasticity and Resistance of the material under consideration.

$$V = \mathbf{R}\, i \quad \text{and} \quad \text{stress} = \mathbf{k} \times \text{strain}$$

Least square fitting is a way to do this and for that we need *sum of squares of the offset (residual)* of the data points from the curve $f(x)$. Offsets can be either *vertical* or *perpendicular*



Vertical offsets allow uncertainties of data points along x, y-axes to be independent of each other.

Why squares and not absolute values?

Squares permit offsets to be treated as continuous differentiable quantities.

Absolute values result in discontinuous derivatives.

Fit N data points (x_i, y_i, σ_i) to a model $f(x_i; a_1, a_2, \ldots, a_M)$ with M parameters ($M < N$). Least square suggests to minimize squares of weighted offsets with respect to the parameters a_k's

$$\chi^2 = \sum_{i=1}^{N} \left(\frac{y_i - f(x_i; a_1, a_2, \ldots, a_M)}{\sigma_i} \right)^2 \rightarrow \frac{\partial \chi^2}{\partial a_j} = 0$$

Linear regression : model the data points with a straight line

$$f(x) = a_1 + a_2 x \;\Rightarrow\; \chi^2 = \sum_{i=1}^{N} \left(\frac{y_i - a_1 - a_2 x_i}{\sigma_i} \right)^2$$

Minimizing χ^2 w.r.t. a_1, a_2 yields

$$\frac{\partial \chi^2}{\partial a_1} = -2 \sum_{i=1}^{N} \frac{y_i - a_1 - a_2 x_i}{\sigma_i^2} = 0$$

$$\frac{\partial \chi^2}{\partial a_2} = -2 \sum_{i=1}^{N} \frac{x_i (y_i - a_1 - a_2 x_i)}{\sigma_i^2} = 0$$

Introducing a shorthand symbol $\sum_{i=1}^{N} \equiv \sum_i$, we ended up with

$$\sum_i \frac{y_i}{\sigma_i^2} = a_1 \sum_i \frac{1}{\sigma_i^2} + a_2 \sum_i \frac{x_i}{\sigma_i^2} \qquad \mathcal{S}_y = a_1 \mathcal{S} + a_2 \mathcal{S}_x$$

$$\equiv$$

$$\sum_i \frac{x_i y_i}{\sigma_i^2} = a_1 \sum_i \frac{x_i}{\sigma_i^2} + a_2 \sum_i \frac{x_i^2}{\sigma_i^2} \qquad \mathcal{S}_{xy} = a_1 \mathcal{S}_x + a_2 \mathcal{S}_{xx}$$

If the data errors σ_i are not given, take $\sigma_i = 1$. However, your code must include σ_i in case they are provided.

The solution of these equations are absolutely straight forward,

$$a_1 = \frac{\mathcal{S}_{xx}\mathcal{S}_y - \mathcal{S}_x \mathcal{S}_{xy}}{\Delta}, \quad a_2 = \frac{\mathcal{S}_{xy}\mathcal{S} - \mathcal{S}_x \mathcal{S}_y}{\Delta} \text{ where } \Delta = \mathcal{S}\mathcal{S}_{xx} - \mathcal{S}_x^2$$

Estimating the errors σ_{a_i} on the parameters a_i determined above

$$\sigma_{a_i}^2 = \sum_i \sigma_i^2 \left(\frac{\partial a_i}{\partial y_i} \right)^2$$

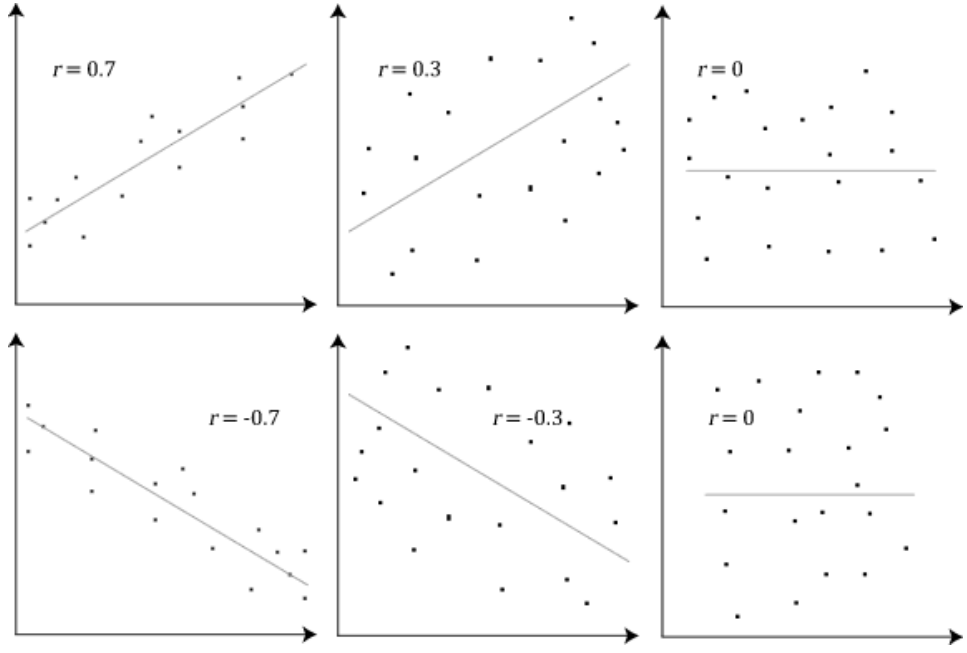
$$\Rightarrow \qquad \frac{\partial a_1}{\partial y_i} = \frac{\mathcal{S}_{xx} - \mathcal{S}_x x_i}{\Delta \, \sigma_i^2}, \quad \frac{\partial a_2}{\partial y_i} = \frac{\mathcal{S} x_i - \mathcal{S}_x}{\Delta \, \sigma_i^2}$$

$$\Rightarrow \qquad \sigma_{a_1}^2 = \frac{\mathcal{S}_{xx}}{\Delta} \text{ and } \sigma_{a_2}^2 = \frac{\mathcal{S}}{\Delta}$$

To this end, in order to get some idea about how *good* is the **linear** fit we define an estimate called *Pearson's correlation coefficient* r

$$r^2 = \frac{\mathcal{S}_{xy}^2}{\mathcal{S}_{xx}\mathcal{S}_{yy}} \quad \text{where} \quad 0 \leq r^2 \leq 1$$

As $r^2 \to 1$ fit gets better.



NB : But is the straight line model itself good to fit the data? To address this requires ideas of *goodness of fit*, *confidence level*, testing some *null hypothesis* against χ^2 probability distribution and so on.

The straight line model is generic enough for use in a few other models which can be reduced to straight line form usually by taking logarithms,

$$\begin{aligned}
\text{exponential :} \quad & f(x) = a\, e^{b\,x} \quad \rightarrow \quad \log f(x) = \log a + bx \\
\text{logarithm :} \quad & f(x) = a + b\, \log x \\
\text{power law :} \quad & f(x) = a\, x^{b} \quad \rightarrow \quad \log f(x) = \log a + b\, \log x
\end{aligned}$$

Least square polynomial fitting

Polynomial model $f(x) = \sum_{k=0}^{n} a_k x^k$ can also be subjected to linear fitting. For simplified expressions, we put all the errors $\sigma_i^2 = 1$ without any loss of generality. In such case, $\chi^2 \rightarrow R^2$

$$\begin{aligned}
R^2 &= \sum_{i=1}^{n} \left[y_i - \left(a_0 + a_1 x_i + \cdots + a_k x_i^k \right) \right]^2 \\
\frac{\partial R^2}{\partial a_0} &= -2 \sum \left[y_i - \left(a_0 + a_1 x_i + \cdots + a_k x_i^k \right) \right] = 0 \\
\frac{\partial R^2}{\partial a_1} &= -2 \sum \left[y_i - \left(a_0 + a_1 x_i + \cdots + a_k x_i^k \right) \right] x_i = 0 \\
&\cdots \\
\frac{\partial R^2}{\partial a_k} &= -2 \sum \left[y_i - \left(a_0 + a_1 x_i + \cdots + a_k x_i^k \right) \right] x_i^k = 0
\end{aligned}$$

Set of equations for a_i's are,

$$\begin{aligned} a_0 n + a_1 \sum x_i + \cdots + a_k \sum x_i^k &= \sum y_i \\ a_0 \sum x_i + a_1 \sum x_i^2 + \cdots + a_k \sum x_i^{k+1} &= \sum x_i y_i \\ &\cdots \\ a_0 \sum x_i^k + a_1 \sum x_i^{k+1} + \cdots + a_k \sum x_i^{2k} &= \sum x_i^k y_i \end{aligned}$$

which when written in matrix form becomes a problem of matrix inversion, and you are free to use your favorite inverter.

$$\begin{pmatrix} n & \sum x_i & \cdots & \sum x_i^k \\ \sum x_i & \sum x_i^2 & \cdots & \sum x_i^{k+1} \\ \vdots & \vdots & \ddots & \vdots \\ \sum x_i^k & \sum x_i^{k+1} & \cdots & \sum x_i^{2k} \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_k \end{pmatrix} = \begin{pmatrix} \sum y_i \\ \sum x_i y_i \\ \vdots \\ \sum x_i^k y_i \end{pmatrix}$$

$$\Rightarrow \quad \mathbf{X} \cdot \mathbf{a} = \mathbf{Y} \quad \rightarrow \quad \mathbf{a} = \mathbf{X}^{-1}\mathbf{Y}$$

Fitting with orthogonal polynomial

Use of orthogonal basis functions to void numerical stability and oscillations. They minimize the maximum error between polynomial and target data over the interval. Because of orthogonality, the regression matrix is well-conditioned, preventing round-off and ill-conditionings. Coefficients are independent of each other

Polynomial	Domain	Weight $w(x)$
Legendre P_n	$[-1, 1]$	1
Hermite H_n	$(-\infty, \infty)$	$\exp(-x^2)$
Laguerre L_n	$[0, \infty)$	$\exp(-x)$

DIY fitting

1. A time versus angular velocity data set on deceleration of a rotating disc is in the file `fit1.dat`. Fit it with the following functions (*i*) $\omega(t) = \omega_0 + \omega_c t$ and (*ii*) $\omega(t) = \omega_0\, e^{-\omega_c t}$. Determine ω_0, ω_c and the quality of fit for both the functions by *Pearson's* r.
2. Distance (r) versus height (h) of the trajectory of a test missile is given in the datafile `fit2.dat`. Try quadratic fit $h = a_0 + a_1 r + a_2 r^2$ and determine the highest point reached by the missile.